Mukhodobwane Vhutalihandivho

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PROFESSIONAL SUMMARY

Final-year BSc Computer Science and Informatics student at the University of Johannesburg, with hands-on experience building full-stack web applications, implementing secure authentication systems, and applying data structures and algorithms to solve real-world problems. Currently mentoring junior CS students through a formal bursary programme, and co-founding a freelance web development partnership delivering production software for real clients. Driven by a genuine hunger to grow through structured, high-intensity environments.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), Java, C++, C#, SQL

Web & Backend: Node.js, Express.js, REST API design, React, Vite, HTML5, CSS3

Databases: PostgreSQL, SQL schema design, relational data modelling

Security: JWT (dual-token with refresh queue), bcrypt, HTTP-only cookies, MFA via email OTP

Tools & Practices: Git, GitHub, OOP, Data Structures & Algorithms, Agile/SDLC, Railway, Netlify

EXPERIENCE

StudyTrust Feb 2026 – Nov 2026

Peer Mentor | Formal Bursary Programme

- Mentored 2 junior Computer Science students through coursework challenges, study planning and academic development as part of a structured bursary programme
- Developed clear, structured explanations of complex topics — data structures, algorithms and OOP — reinforcing my own understanding while supporting peers
- Assigned an industry mentor through the programme, gaining exposure to professional engineering practices and career planning in a real-world context

Elevate Digital Solutions (EDS) Jan 2026 – Present

Co-Founder & Technical Lead | Freelance Web Development Partnership

- Co-founded a freelance web development partnership while completing final year of studies — took on real clients, managed projects end-to-end and delivered production software outside of academic work
- Led technical delivery across projects: scoping requirements, architecting solutions, writing production code and handling client communication
- Delivered production websites for clients including an NPO (Thusa Lushaka Empowerment), handling ongoing maintenance and post-launch support

PROJECTS

MutualTrust – Student Management System (KSMS) Jan 2026 – Oct 2026

Final Year Project | React, Node.js, PostgreSQL

- Built both frontend and backend of a multi-role platform serving student, teacher, centre manager and admin workflows — full ownership of the authentication and API layers
- Implemented secure auth flows including dual-token JWT, HTTP-only cookies, MFA via email OTP and a refresh queue to handle concurrent API calls safely
- Developed shared utility modules standardising API communication, validation and UI behaviour across all role-based dashboards
- Collaborated in a team using Git branching, code reviews and integrating independently developed modules — applying Agile practices throughout

Fingerprint Recognition System Apr 2026 – May 2026

Final Year Project | Java, Graph Algorithms

- Designed a graph-based fingerprint matching pipeline using Zhang-Suen skeletonisation to extract ridge structure, with minutiae points represented as graph nodes
- Implemented k-NN graph matching to compare fingerprint graphs, enabling similarity scoring and identity classification across benchmark datasets (FVC2002, FVC2004)
- Applied graph theory concepts — adjacency, traversal, k-NN — to a real-world biometric problem, bridging academic data structures work with practical application

Thusa Lushaka Empowerment – NPO Website 2026
Client Project (EDS) | React, Vite, React Router, Netlify

- Designed and built a complete multi-section website for a South African youth NPO (Thusa Lushaka Empowerment, Nzhelele, Limpopo) — covering their programmes, annual academic awards, and the Miss Thusa Lushaka beauty pageant
- Built dedicated pages and sections for About, Programs, Awards, Pageant, Gallery and Donate, with React Router handling client-side navigation and Netlify Forms powering the contact and inquiry forms without a backend
- Took the initiative to build and deploy the site proactively, then pitched it to the organisation — handling the full client conversation from first email through to demo presentation

Job Application Tracker 2025
Full-Stack Web App | Node.js, Express.js, PostgreSQL, JWT | Deployed to Production

- Built a full-stack REST API backend in Node.js/Express deployed on Railway, with a plain HTML/CSS/JS frontend hosted on Netlify — live and publicly accessible
- Implemented JWT authentication with bcrypt password hashing, protecting all job-related endpoints from unauthenticated access
- Supported full CRUD for job applications with status tracking across four stages — Applied, Interview, Offer and Rejected — giving users a clear pipeline view of their search
- Resolved cross-origin resource sharing (CORS) issues between separate frontend and backend deployments, a common production challenge for full-stack apps

Interactive 3D Developer Portfolio 2026
Personal Project | React, Vite, React Three Fiber (Three.js) | Live Deployed

- Built a dual-mode portfolio: a clean professional page for recruiters and an interactive 3D "hacker room" (React Three Fiber) where visitors sit at a virtual desk and explore my CV on the monitors
- The 3D room features a typed terminal, a fingerprint-scan boot sequence, a tech trivia quiz and a password-cracking mini-game — with a text-only fallback for low-powered devices and tap controls for mobile
- 190 unit tests (Vitest) and 17 end-to-end tests (Playwright); Lighthouse 100 for performance, accessibility and SEO; Netlify Forms contact with spam protection; CSP headers and zero cookies — live at mukhodobwanevhutalihandivho.netlify.app

EDUCATION

University of Johannesburg 2023 – 2026
BSc Computer Science and Informatics

- Relevant coursework: Data Structures & Algorithms, Software Engineering (Requirements, SDLC, Agile), Priority Queues, Heaps, Hash Tables, Graph Theory, Binary Trees
- Applied coursework practically through team and individual projects spanning backend APIs, graph-based systems and secure authentication architectures
- South African citizen | Based in Johannesburg